MSDI Calculator

Capital Cost Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

Prepared by: Nathan Sparrow

Prepared date: July 8, 2026

Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: *Methodology documentation for screening-level capital cost calculations based on current app logic and 2025 quote-derived pricing assumptions.*

Purpose and Scope

This technical memorandum documents the capital cost methodology used by the MSDI Calculator to estimate the installed capital expenditure associated with converting eligible dairy acreage to a manure subsurface drip irrigation (mSDI-e) system. The methodology describes the cost categories, source pricing basis, scaling equations, salvage-value treatment, cost-share treatment, and annualized capital-cost logic used for screening-level model outputs.

The methodology is intended for technical documentation, model review, and future consolidation into a broader MSDI Calculator methodology manual. It is not intended to replace site-specific engineering design, vendor procurement, tax advice, or formal financial underwriting. Actual installed costs may vary materially depending on site layout, water quality, solids separation requirements, field shape, elevation, electrical service, pipe routing, trenching conditions, and local labor and material costs.

Terminology and Units

Term	Definition in this methodology	Primary unit
Acres for SDI	User-entered acreage proposed for manure subsurface drip irrigation conversion.	acres
Pump station	Turnkey pump-station component package and associated installation cost used to deliver water/manure blend to the mSDI system.	$/station
Filter station	Turnkey filtration component package and associated installation cost used to protect the drip system from clogging.	$/station
Maximum acres per station	Reference design capacity for one pump/filter station; current methodology uses 160 acres per station.	acres/station
Drip tape and pipe	In-field drip hose/submains/in-field valves plus installation/trenching and mainline pipe costs.	$
Mainline distance	Combined user-entered distance from manure source to station, water source to station, and station to fields.	feet
Salvage value	User-entered value of retired equipment displaced by the mSDI project.	$
Simple annual amortized cost	Screening-level straight-line annual cost equal to installed component cost divided by useful life.	$/year

Source Pricing Data and Cost-Basis Assumptions

Capital-cost coefficients are based on a 2025 vendor quote for a California dairy mSDI installation. The quote-derived values are used as nationally applicable screening-level planning assumptions. This does not imply that California cost conditions are identical to all U.S. dairy regions; rather, the quote provides a current, real-world installed-system basis suitable for preliminary feasibility screening.

The capital-cost lookup separates material/component costs from installation costs where the model benefits from separate scaling. Pump and filter stations are treated as turnkey station-level cost categories, while drip tape, trenching, and mainline costs are scaled by acreage or distance. This structure preserves the practical distinction between fixed station costs, acreage-proportional field costs, and distance-proportional conveyance costs.

Cost category	Scaling basis	Interpretation
Pump station + pump station installation	Station count	Fixed station-level cost; one station supports up to 160 acres.
Filter station + filter station installation	Station count	Fixed station-level cost; one station supports up to 160 acres.
Drip hose, submains, in-field valves	Acres	Field-level distribution system material cost.
Field installation / trenching	Acres	Acreage-proportional installation cost for in-field drip placement.
Mainline + mainline installation	Feet	Distance-proportional conveyance cost for water, manure, and field routing.
Salvage value of retired equipment	User input	Offset representing value of displaced/retired equipment.

Pump and Filter Station Sizing

The model treats pump and filter stations as discrete infrastructure units. A single pump/filter station is assumed to support up to 160 acres of mSDI-e service. This is a sizing limit, not a continuously variable acreage cost. Therefore, costs do not scale linearly from acre 1 to acre 160. A 95-acre system and a 160-acre system both require one station, while a 200-acre system requires two stations.

Equation 1. N_station = CEILING(A_SDI / A_station,max)

Where N_station is the number of required pump or filter stations, A_SDI is proposed mSDI acreage, and A_station,max is the maximum acreage served by one station. The current model basis is 160 acres per station.

The same station-count logic is used for pump components and filter components. The implementation applies the ceiling function to the ratio of proposed mSDI acreage to maximum station capacity, so partial station requirements are rounded upward to the next whole station.

Pump Component Cost

Pump component cost is calculated as the number of required stations multiplied by the combined pump-station material/component cost and pump-station installation cost. The calculation uses lookup-table values for pump station cost and pump station installation cost.

Equation 2. C_pump = N_station x (C_pump_station + C_pump_install)

Pump station costs are treated as fixed station-level costs. The model does not interpolate partial pump station cost below the 160-acre reference capacity.

Filter Component Cost

Filter component cost uses the same station-count approach as pump components. Filtration is treated as a required station-level component for manure SDI because solids and clogging risk must be controlled before manure/liquid reaches the drip system.

Equation 3. C_filter = N_station x (C_filter_station + C_filter_install)

Filter station costs are treated as fixed station-level costs. The selected coefficient includes the filter station package and associated installation basis.

Drip Tape, In-Field Pipe, and Mainline Cost

The drip tape and pipe category combines three distinct cost mechanisms: acreage-proportional in-field installation, acreage-proportional drip material cost, and distance-proportional mainline conveyance cost. This reflects the fact that in-field drip infrastructure scales primarily with treated acreage, while mainline cost scales with the physical distance between manure source, water source, station, and fields.

Equation 4. $\text{C\_trench} = \text{C\_trench,ac} \times \text{A\_SDI}$

Equation 5. $\text{C\_drip} = \text{C\_drip,ac} \times \text{A\_SDI}$

Equation 6. $\text{L\_mainline} = \text{D\_manure\_to\_station} + \text{D\_water\_to\_station} + \text{D\_station\_to\_fields}$

Equation 7. $\text{C\_mainline} = \text{L\_mainline} \times (\text{C\_mainline,ft} + \text{C\_mainline\_install,ft})$

Equation 8. $\text{C\_drip\_pipe\_total} = \text{C\_trench} + \text{C\_drip} + \text{C\_mainline}$

This structure is directly consistent with the app implementation: trenching cost and drip hose cost are multiplied by acres for SDI, while total pipe distance is multiplied by the combined mainline material and installation rate.

Salvage Value

The model allows the user to enter the value of retired equipment displaced by the mSDI project. This value is treated as an offset to total project cost or as an annualized offset depending on the reporting context. The purpose is to avoid overstating net capital burden when existing equipment has recoverable resale, trade-in, or avoided replacement value.

Equation 9. $\text{C\_salvage} = \text{User-entered retired equipment value}$

Equation 10. $\text{C\_project} = \text{C\_pump} + \text{C\_filter} + \text{C\_drip\_pipe\_total} - \text{C\_salvage}$

Total Installed Capital Cost

Total installed capital cost is calculated by summing pump components, filter components, and drip tape/pipe components, then subtracting the user-entered salvage value of retired equipment. This produces the screening-level project capital cost before cost-share treatment.

Equation 11. $\text{C\_total\_without\_cost\_share} = \text{C\_pump} + \text{C\_filter} + \text{C\_drip\_pipe\_total} - \text{C\_salvage}$

The model output may also display subtotal lines for pump components, filter components, drip tape and pipe, salvage value of retired equipment, and total cost without cost-share. These subtotals are intended to make the calculation transparent to the user.

Cost-Share Treatment

The MSDI Calculator includes cost-share parameters so that project costs can be evaluated both with and without financial assistance. The model uses user-entered cost-share percentage and payment-cap assumptions to estimate the farmer-paid portion of the capital cost. When a maximum payment cap is supplied, the subsidy amount is limited by that cap.

Equation 12. $\text{Subsidy\_intended} = \text{C\_total\_without\_cost\_share} \times \text{CostShare\_rate}$

Equation 13. $\text{Subsidy\_actual} = \text{MIN}(\text{Subsidy\_intended}, \text{Subsidy\_cap})$

Equation 14. $\text{C\_farmer\_paid} = \text{C\_total\_without\_cost\_share} - \text{Subsidy\_actual}$

This treatment is intended for screening-level feasibility analysis. Actual cost-share eligibility, payment limits, ranking criteria, practice standards, and payment timing are program-specific and must be verified independently.

Simple Annual Amortized Cost

The current results methodology uses a simple straight-line annualization approach to express installed capital cost as an annual screening-level cost. This approach divides each major capital component by its useful life. The calculation is intended to approximate simple annual ownership burden for screening comparison; it is not a tax depreciation schedule, loan amortization schedule, or discounted cash-flow analysis.

Equation 15. AnnualCost_i = C_i / UsefulLife_i

Component	Annualization basis	Interpretation
Pump components	Pump component cost / pump useful life	Annualized pump ownership cost.
Filter components	Filter component cost / filter useful life	Annualized filter ownership cost.
Drip tape and pipe	Drip tape and pipe cost / drip-field useful life	Annualized in-field system ownership cost.
Salvage value of retired equipment	Applied as an offset in project-cost and/or annualized reporting context	Recognizes recoverable value of displaced existing equipment.

Equation 16. AnnualCapitalCost = SUM(AnnualCost_i) - AnnualizedSalvageOffset

The simple annualized cost approach is consistent with the calculator's screening-level purpose. IRS Publication 946 explains that depreciation generally recovers the cost of income-producing property over time; however, the MSDI Calculator does not attempt to calculate tax depreciation, Section 179 deductions, bonus depreciation, or MACRS tax treatment. Instead, the model reports a transparent straight-line annualized planning value.

Relationship to Results Outputs

Capital-cost outputs support two main result areas: the capital expenditure table and the operational expenditure comparison. The capital expenditure table presents installed cost components and simple annualized cost values. The operational expenditure comparison uses the annual capital cost as the capital-cost line item contributing to annual net expense or benefit.

Output element	Capital-cost source
Pump Components	Pump station cost equation.
Filter Components	Filter station cost equation.
Drip Tape and Pipe	Acreage-based drip/trenching plus distance-based mainline equation.
Salvage Value of Retired Equipment	User-entered retired equipment value.
Total Without Cost-Share	Pump + filter + drip/pipe - salvage.
Project Total With Cost-Share	Total cost adjusted for user-entered cost-share rate and cap.
Capital Costs (Simple Amortized)	Straight-line annualized project cost used in annual results.

Assumptions and Limitations

- Capital prices are based on a 2025 California dairy mSDI quote which was update for 2026 according to conversations with the Irrigation Materials Vendor and Field Installer and are used as nationally applicable screening-level values.
- Pump and filter stations are sized discretely in 160-acre increments; the calculator does not interpolate partial station costs below the reference capacity.
- Pump and filter station values are treated as turnkey/inclusive costs for the represented component category.
- The model does not perform hydraulic design, pump curve selection, pressure-loss modeling, or filtration sizing based on water quality.
- Mainline cost is estimated from user-entered distance values and does not account for elevation, road crossings, utility conflicts, easements, permitting, or special trenching conditions.

- Straight-line annualization is used for screening and does not include debt service, taxes, discounting, depreciation recapture, inflation, or opportunity cost of capital.
- Cost-share calculations are simplified and should not be interpreted as a guarantee of program eligibility or payment amount.

Quality Control and Validation Checks

- Verify that Cost_LU values are expressed in 2025 dollars and match the latest approved pricing source.
- Confirm that maximum acres per station remains consistent with the pump/filter design basis.
- Test acreage breakpoints immediately below and above 160 acres to confirm station-count rounding behavior.
- Test distance fields independently to verify mainline cost scales only with distance and not acreage.
- Confirm that the displayed capital expenditure table matches pump, filter, drip/pipe, salvage, total, and cost-share calculations.
- Confirm that the annualized capital-cost line item matches the simple straight-line annualization methodology used in current outputs.

Processing Workflow Summary

1. Read proposed mSDI acreage from user inputs.

2. Calculate number of pump stations using the 160-acre station-capacity rule.

3. Calculate pump component cost using station count, pump station cost, and pump installation cost.

4. Calculate filter component cost using station count, filter station cost, and filter installation cost.

5. Calculate in-field trenching and drip hose costs using acreage-based rates.

6. Calculate mainline cost using combined manure, water, and field distance multiplied by material-plus-installation cost per foot.

7. Sum pump, filter, and drip/pipe costs and subtract user-entered salvage value.

8. Apply cost-share rate and cap where applicable.

9. Calculate simple annualized capital cost using useful-life assumptions.

10. Pass capital-cost outputs to the capital expenditure table and annual expense-change results.

References and Source Links

Project-specific 2025 mSDI capital-cost quote. California dairy mSDI installation quote used as the primary pricing basis for pump station, filter station, drip tape, mainline, and installation costs. Internal project source.

USDA NRCS Conservation Practice Standard: Irrigation System, Microirrigation (Code 441). https://efotg.sc.egov.usda.gov/references/public/DO/CO441-3-02.pdf

USDA NRCS Conservation Practice Standard: Irrigation System, Surface and Subsurface (Code 443). https://www.nrcs.usda.gov/resources/guides-and-instructions/irrigation-system-surface-and-subsurface-ac-443-conservation

Kansas State Research and Extension. Subsurface Drip Irrigation Components: Minimum Requirements.. https://www.wkrec.org/programs/irrigation_engineering/subsurface_drip/reports/MF2576.pdf

IRS Publication 946. How To Depreciate Property.. https://www.irs.gov/publications/p946

Appendix A. Implementation Traceability

Calculation area	Implementation file/function	Equation basis
Pump components	row-21-pump-components.ts / calculatePumpComponentsCost()	CEILING(acres / max acres per station) x (pump station + pump installation).
Filter components	row-22-filter-components.ts / calculateFilterComponentsCost()	CEILING(acres / max acres per station) x (filter station + filter installation).
Drip tape and pipe	row-23-drip-tape-and-pipe.ts / calculateDripTapeAndPipeCost()	Trenching x acres + drip hose x acres + total distance x (mainline + mainline installation).
Salvage value	row-24-salvage-value.ts / getSalvageValueRetiredEquipment()	User-entered retired equipment value.
Total cost	row-25-total-cost.ts / calculateTotalMsdiCost()	Pump + filter + drip/pipe - salvage.
Annualized cost	Current results methodology	Simple annualized cost = component cost / useful life, with salvage/cost-share treatment as reported in outputs.